The Integration of Digital Technology in the Innovation Management of Enterprise Innovation Ecosystems

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Abstract

The paper researches that an enterprise improves innovation management using digital technologies by constructing the digital innovation ecosystem. It delves into three aspects, namely, industrial chain collaborative innovation mechanisms, opening and sharing mechanisms of the innovation chain, and the integration mechanisms of the industrial chain and innovation chain. The study suggests that effective relationship mechanisms are the key to building an enterprise's innovation ecosystem, which can promote continuous innovation and development. The industrial chain collaborative innovation mechanism includes aspects such as "Internet+" marketing collaborative innovation, collaborative production and manufacturing data, data-driven technological collaborative innovation, and collaborative construction of an innovative ecosystem within the industry. The opening and sharing mechanism of the innovation chain encompasses multi-dimensional resource-sharing mechanisms and the construction of a digital open innovation platform. Furthermore, the paper analyzes the integration mechanisms of the industrial chain and innovation chain, including Independent innovation strengthen the support system, Digital innovation to create a dynamic system, Organizational innovation and upgrading capability system. The continuous development and mutual integration of the innovation chain and industrial chain effectively facilitate the flow and exchange of internal and external innovation resources, jointly driving digital transformation and innovation development.

CCS Concepts

• **Information systems** → Information systems applications; Enterprise information systems.

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Keywords

Digital innovation ecosystem, Relationship mechanisms, Collaborative innovation mechanisms, Opening and sharing mechanisms, Integration mechanisms

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1 Introduction

In recent years, the widespread application of digital technologies has changed the production modes and business models of traditional industries. The digital economy has become a new driving force for economic recovery and growth. China's digital economy is developing particularly rapidly. According to the Digital China Development Report (2023), the scale of online retail sales was 15.42 trillion yuan in 2023 and the added value of core industries in the digital economy accounted for about 10% of GDP. Specifically, on the one hand, under the wave of digital industrialization, emerging digital technologies and business models continue to emerge. Many digital innovation platforms, as the dominant player in the innovation ecosystem, have become an important force in promoting social and economic development[1]. On the other hand, industrial digitization is also accelerating. A large number of traditional industries are using the innovation ecosystem to carry out digital transformation, which fuels the continuous emergence of new modes and new forms of business, such as industrial Internet and intelligent manufacturing[2].

In the context of digital transformation, digital innovation is gradually being integrated into the enterprise innovation ecosystem, forming a digital innovation ecosystem. And the enterprise digital innovation ecosystem has become a new mode of enterprise innovation[3]. The rapid development of the digital economy and the widespread application of digital technologies have positioned data as a new factor of production, leading to novel combinations of factors. At the same time, digital technologies have also facilitated the optimized reorganization of existing production factors[4], providing favorable opportunities and conditions for the construction and operation of an enterprise's digital innovation ecosystem. However, the heterogeneity among innovation entities and the complexity of competing relationships in the context of the digital economy also brings new problems and challenges to the development of the enterprise digital innovation ecosystem. On the one

hand, the main body of enterprise digital innovation ecosystem organization includes all enterprises, organizations, individuals, and corresponding environmental factors related to their interests. As digital technologies proliferate and permeate various sectors, the boundaries of entities within the systemic organization are disrupted, transforming the enterprise into an open platform. On this platform, small groups composed of individuals emerge, and existing individuals coalesce to form innovative communities, with even single individuals emerging as innovative entities. Furthermore, under the influence of digital technology applications, any enterprise or individual can participate in the entire process from idea generation to technology diffusion anytime, anywhere. This trend underscores the democratization of innovation, the blurring of status boundaries, the equalization of relationships between enterprises, and the liberalization of entry and exit.

The paper adopts a case study method and selects China Tobacco Guangxi Industrial Co., Ltd. (hereinafter referred to as China Tobacco Guangxi) for analysis. The company has embarked on the construction of digital R&D and digital raw material platforms, clarified five major goals for the development of digital R&D innovation platforms, formulated a "1642" overall architecture for raw material digitization, conducted digitalization work for tobacco leaf and module formulations, and established a brand consumer migration model. China Tobacco Guangxi has made active explorations in digital innovation and achieved certain results. However, it also faces new challenges in integrating internal and external resources to better promote digital transformation and technological innovation. This paper aims to delve into the basic structure, relational mechanisms, and practical applications and implications of enterprise digital innovation ecosystems through a case study of China Tobacco Guangxi.

2 Theoretical Review

The theory of innovation ecosystems originates from the concept of business ecosystems. It refers to a collaborative network built around corporate innovation activities. Through cooperative arrangements, participants combine their respective product portfolios to form a coherent, customer-oriented solution[5]. The innovation ecosystem is also a dynamic, networked, multi-stakeholder collaborative system that promotes the transformation of innovative ideas into innovative products and fully taps into the innovative potential of enterprises[6-7]. It emphasizes that participants achieve innovation through interactive coordination, thereby jointly creating value for customers. Based on existing research, its connotation mainly includes the following three aspects. Firstly, structurally, the innovation ecosystem functions as an ecological organization[8], characterized by complex networks with nonlinear interactive relationships such as symbiotic coupling and competitive cooperation between entities[9-10]. Secondly, from the perspective of external environmental relationships, the innovation ecosystem is an open system where innovative entities within the system interact with the external innovation environment. This environment provides resource support for innovative entities and their behaviors[11]. Thirdly, functionally, the innovation ecosystem is a sustainable system. This sustainability is reflected in the diversity and continuity of innovation activities, as well as the continuous evolution of the system itself [12].

Digital transformation is an emerging concept that has arisen with the development of the digital economy [13]. It refers to the process of utilizing digital technology to drive changes and innovations in corporate production methods and business processes[14]. The application of digital technology has reshaped the way innovative entities co-create value, promoted the development of innovation ecosystems, and given rise to a new model of digital innovation ecosystems[15]. Specifically, this shift marks a transition from enterprise-centered value creation to interactive value co-creation among multiple actors within the innovation ecosystem[16]. The innovation ecosystem offers a dynamic evolutionary perspective for studying value creation in the digital context, emphasizing the roles of multiple actors, their innovative behaviors, and their interactions[17-18]. A digital innovation ecosystem is a complex economic structure established by interacting organizations and individuals, relying on digital technology to promote collaborative innovation in products and services[19-20]. It has specificities in terms of innovation elements, participating entities, and inter-entity relationships[21]. Firstly, innovation factors become more digitized. A major distinction between digital innovation ecosystems and other types of innovation ecosystems lies in the circulation of data elements[22], which serves as a manifestation of innovation ecosystems in a digital context. This can be understood as a fusion of the concepts of digital innovation and innovation ecosystems[23]. In this context, digital resources, digital technology, and digital infrastructure serve as key elements of the innovation ecosystem, playing a fundamental and transformative role in the innovation process[24]. Secondly, participating entities become more virtualized. Innovation entities exhibit greater heterogeneity and the flow of resources is more complex[25]. Based on digital technology, the breadth and flexibility of connections between participants have significantly increased. A vast number of highly heterogeneous participants become the mainstay of value creation, and innovative entities rely on platforms to achieve boundary less virtual links[26]. Thirdly, the relationship becomes more ecological. The programmability and data homogeneity of digital technologies have led to an increasing blurring of the boundaries between organizations and their environments, resulting in a more dynamic and open relationship between participants, and a change in innovation collaboration from a sequential linear supply chain model to a multi-layered and networked ecosystem model[27-28].

Based on the theory of innovation ecosystems, the enterprise digital innovation ecosystem further highlights the prominent role of enterprises driven by digital technology, as well as the synergistic effects of the innovation process and results. Current domestic research focuses on digital technology, digital connectivity, and innovation collaboration among participants[3]. It points out that digital transformation requires a shift in governance concepts and models, breaking down information islands and achieving comprehensive information access[29]. Through innovative governance measures such as the disclosure, integration, and application of information resources, the ecosystem's innovation capability can be enhanced. Some other scholars have researched the resilience of innovation ecosystems and digital innovation ecosystems[30-31].

It can be seen that theoretical research on digital transformation and innovation ecosystems has achieved certain results. However, integrated research that bridges these two domains remains scant. Especially in specific scenarios, there is a lack of in-depth research on the relationship mechanisms constructed by entities within the enterprise digital innovation ecosystem based on their functional positioning.

3 Research Design

3.1 Research Methodology

This paper primarily delves into the question of how the constituent entities and their relationship mechanisms within an enterprise's digital innovation ecosystem play a role in the company's digital transformation and innovation. While some studies have focused on digital transformation or innovation in an enterprise's digital innovation ecosystem, there is a lack of research on the relationship mechanisms of digital innovation ecosystems in tobacco industry enterprises, and corresponding theoretical models have not been established. Therefore, it is difficult to adopt a quantitative empirical research method by directly proposing theoretical hypotheses. Case study methodology serves as a research strategy that aims to describe, explain, explore, or evaluate phenomena, generalizing, and summarizing the case through a rigorous research design under the guidance of research questions and theories, and by comprehensively applying the methods of data collection and analysis[32], thereby rendering it suitable for addressing research questions such as "how to do it" "why" and "what are the results"[30]. Therefore, this paper opts for an exploratory case study approach. Meanwhile, to ensure the integrity of the data chain and guarantee a thorough analysis of the case, a single-case study method is adopted[33-34] to deeply analyze the composition, functional positioning, and relationship mechanisms of the digital innovation ecosystem of China Tobacco Guangxi.

3.2 Case Selection

The choice of China Tobacco Guangxi as the research object is mainly based on two reasons. Firstly, the case is representative. China Tobacco Guangxi is a large state-owned enterprise and one of the 18 provincial-level cigarette industrial enterprises. It has a professional department to combine digital technology and innovation management. As the only Internet Research Center in the entire tobacco industry, China Tobacco Guangxi has achieved remarkable results in orderly promoting digital transformation and enterprise digital innovation ecosystem work. Secondly, the data and material are available. The research team has been paying long-term attention to the development of the company, enabling a better understanding of the situation of the case and access to relevant materials and documents.

3.3 Data Collection and Analysis

This study adopts a retrospective case analysis method, following the case analysis framework of "phenomenon description - in-depth analysis - revealing the essence"[30]. Taking the digital innovation ecosystem of China Tobacco Guangxi, we first collect relevant information regarding the company's digital transformation, sorting out the active exploration process of digital transformation that the company has undertaken since 2013. Secondly, based on this, we analyze the constituent entities and functional positioning of the digital innovation ecosystem of China Tobacco Guangxi. Then, we reveal the relationship mechanisms behind the smooth operation of the enterprise's digital innovation ecosystem and answer the question, "What kinds of relationship mechanisms have promoted the development of the digital innovation ecosystem of China Tobacco Guangxi?" By addressing the "What" and "Why" questions, we aim to draw general conclusions and discover the common evolution patterns of enterprise innovation ecosystems.

4 Composition of the digital innovation ecosystem of China Tobacco Guangxi

Enterprise digital innovation ecosystem denotes a collaborative system fostering competitive edge, where enterprises collaborate using digital technology to enhance innovative products and services, and realize collaborative innovation and value co-creation[3]. This ecosystem is characterized by openness, heterogeneity, dynamism, and symbiosis. During the tobacco industry's digital transformation, the interplay has changed among industrial enterprises, commercial entities, cigarette retailers, and consumers. The traditional ecosystem fails to emphasize the heterogeneity of innovation subjects, and the participants of heterogeneous virtual entities complicate innovation subject's coordination. Since 2013, China Tobacco Guangxi has explored "Internet +" integration and innovation, facilitating digital transformation and upgrading. Its digital transformation can be divided into four stages, as detailed in Table 1.

Over a decade of digital transformation efforts, China Tobacco Guangxi has cultivated a robust enterprise digital innovation ecosystem, as shown in Figure 1, which consists of the integration of the industrial chain and the innovation chain. In this system, bidirectional arrows represent the flow of knowledge and information, while unidirectional arrows indicate the influence of policies and digital technologies.The tobacco industry's entire value chain revolves around cigarette products, encompassing all stakeholders involved in production and sales services. The China Tobacco Guangxi industry chain is a structured ensemble comprising raw material suppliers, equipment providers, industrial enterprises, commercial entities, cigarette retailers, and consumers. It contains two main features. Firstly, it is operated under tobacco specific laws and regulations. Secondly, owing to the vertical management and macro-guidance, various segments and localities within the industry chain maintain a relatively balanced state influenced by the current planning system, fiscal and tax system, and stakeholders' interests[35]. The innovation chain of China Tobacco Guangxi comprises other industrial enterprises, equity participation and holding companies, and outsourcing partners (e.g., colleges and universities, scientific research institutes, and intermediary organizations). The innovation chain provides all-round innovation support and resource sharing. Digital technology further strengthens cooperation and synergies within the innovation chain.

Table 1: The Four Stages of Digital Transformation at China Tobacco Guangxi.

Year	Period	Technology	Application	Innovation
2013-2014	Exploration Stage	QR code technology	Marketing	Opening up a new stage of consumer-oriented marketing
2015-2016	Expansion Stage	Radio Frequency Identification	Warehousing Network procurement	Realize lean management and reduce the operating costs of each link
2017-2018	Collaboration stage	Data mining Data Visualization Technology Convolutional neural network	Enterprise management Employee training	Creating an industrial collaborative innovation mode of ”industrial-commercial collaboration, data sharing, one-party development, and multi-party use ”
2019-present	Empowerment stage	Near infrared technology Cloud computing Technology Internet of Things	Raw materials Digital R&D Manufacture	Accelerate the deep integration of digital technology and business and set up the ”Internet+” joint laboratory with other partners in the industry

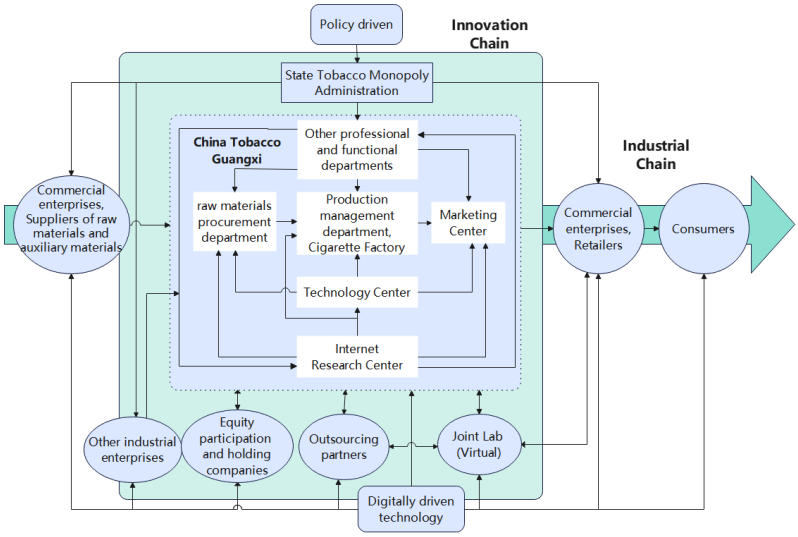

Figure 1: Schematic Diagram of the Digital Innovation Ecosystem Structure of China Tobacco Guangxi.

5 The Relationship Mechanism of the Digital Innovation Ecosystem of China Tobacco Guangxi

The relationship mechanism refers to the way of contact and interaction between the enterprise and other related subjects. The relationship mechanism among various participants determines the level of their cooperation, degree of information sharing, and modes of resource exchange. Based on the analysis of the main subjects and functional positioning of the industrial chain and innovation chain in the digital innovation ecosystem of China Tobacco Guangxi, the relationship mechanism is also explored from the perspectives of the industrial chain collaborative innovation mechanisms, opening and sharing mechanisms of the innovation chain, and the integration mechanisms of industrial chain and innovation chain.

5.1 Industry chain collaborative innovation mechanisms

Collaborative innovation is the result of knowledge and technology integration of different organizations, and inter-organizational relationships affect the interaction and complementary effects of innovation[36]. The operation of collaborative innovation requires control and guidance of both the intrinsic knowledge activities within each stage and the interlocked knowledge activities between stages[37]. The co-innovation mechanism of the digital innovation ecosystem industry chain of China Tobacco Guangxi integrates data resources through digital technology, forms a closed-loop business,

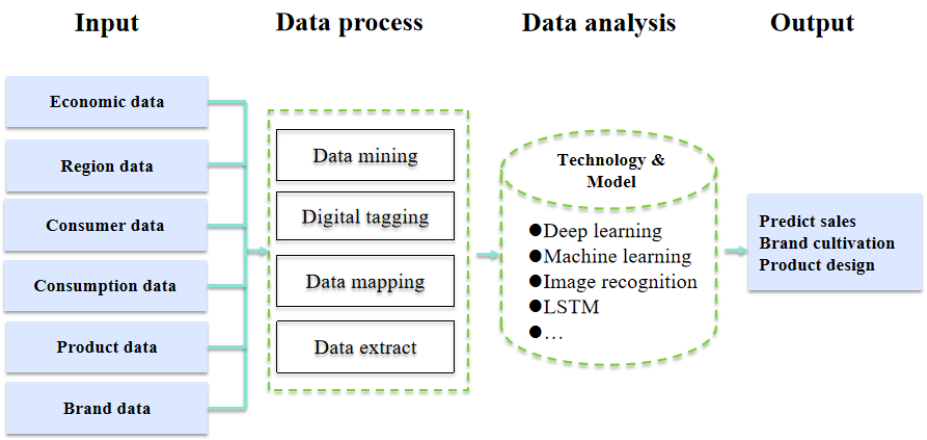

Figure 2: Application Process of Digital Technology in Internet+ Marketing collaborative Innovation.

and comprehensively enhances the level of collaborative interaction among industrial enterprises, commercial entities, cigarette retailers, and consumers. It mainly includes the following four aspects:

5.1.1 "Internet+" marketing collaborative innovation. China Tobacco Guangxi, alongside other partners in the industry, integrates resources and complements strengths, establishing a novel "Internet+" industrial-commercial collaborative marketing model. By collecting sales data, the company has aggregated market data encompassing 53 key cooperative markets, 197,000 urban business districts, and 1.5 million retail terminals. Through integrating big data and AI with tobacco applications, they have studied intelligent cigarette placement, achieving demand forecasting and product placement driven by AI and data analysis, as shown in Figure 2. The company integrates and consolidates multi-source consumption big data, analyzes and researches the characteristics of consumer consumption places, the characteristics of the consumption crowd, and the migration law of consumption, and integrates the mode of "grade + data label" to formulate the strategy of cultivating its "Zhenlong" brand, which effectively improves the precision rate of cargo placement. Through the way of industrial and commercial alliance, Internet innovation and cigarette marketing work in all areas of the depth of integration, to promote information technology upgrading, overall efficiency, organizational management innovation, and efforts to build a consumer as the core, with "new entry point for data collection, new platform for enterprise management, new model for data analysis, new process for resource allocation, new terminals for retailing of cigarettes, new mode of cooperation between industry and commerce "as the support, to enhance the overall efficiency of cigarette marketing as the goal of the modern tobacco cigarette marketing system.

5.1.2 Collaborative production and manufacturing data. The digital platform enables data sharing and application across production processes, optimizing information flow and decision-making, enhancing production efficiency, quality assurance, marketing precision, and fostering collaborative innovation in manufacturing. China Tobacco Guangxi, a pioneer in the tobacco industry, employs "one product, one code" technology on cigarettes. After years of refinement, it has established a comprehensive 2D code system supporting its "Zhenlong" brand. Leveraging QR code technology and adhering to national policies, China Tobacco Guangxi has researched and implemented projects related to cigarette small box, strip box, carton cigarette, and retailer order linkage based on QR code, to achieve full-process traceability from production to consumption for every pack of cigarettes.

5.1.3 Data-driven technology collaborative innovation. China Tobacco Guangxi has undertaken numerous technological innovation researches by leveraging digital technology. As digital research and development efforts have deepened, product development efficiency has seen significant improvement. In recent years, this has resulted in the development of three innovative products and the completion of over ten product quality improvement projects. Additionally, two brand products have adopted formula maintenance and auxiliary design technology, facilitating the upgrade and substitution of tobacco leaf raw materials.By carrying out the construction of a digital R&D application platform for cigarette products, it has introduced artificial intelligence technology to assist product R&D and design and promoted the transformation of R&D from experience-driven to data-driven. To ensure the rational supply of raw materials, a tobacco quality big data platform has been established, leveraging near-infrared technology and AI to construct models for raw material quality index, raw material style index, and formula module design, thereby enhancing the long-term and short-term supply-demand balance and guarantee capabilities of raw materials. With the continuous deepening of research and digital technology, the market awareness and data awareness of R&D personnel are further enhanced, strengthening the effective connection between product research and development and market demand.

5.1.4 Collaborative construction of an innovative ecosystem within the industry. The collaborative innovation within the industrial chain of China Tobacco Guangxi hinges on the intimate collaboration between enterprises across various nodes. On the one hand, the company has established a digital marketing service platform to

engage key retailers and consumers, fostering a retailer-consumer activity hub. This platform enables dynamic incentives through a star growth system. It also carries out digital marketing activities such as sending maintenance, information maintenance, and star maintenance, and outputting management advice, online question and answer, online recommendations, and other interactive publicity and recommendation activities, to jointly promote the digital transformation of marketing activities. Empowering retailers has enhanced management efficiency, brand cultivation, channel control, terminal service, and customer profitability, propelling its brand's local market expansion. On the other hand, the Internet Research Center at China Tobacco Guangxi stressed the integration of Internet innovations in the tobacco industry, augmenting digitization capabilities and fostering "Internet+" industry-commerce collaborations. Leveraging the "Internet+" joint laboratory, the company has achieved tangible results, establishing a collaborative innovation model of "industrial-commercial collaboration, data sharing, one-party development, and multi-party use". The mode of innovation has shifted from the closed innovation of a single enterprise to the open innovation of industrial-industrial cooperation and industrial-commercial cooperation.

These mechanisms foster close collaboration between China Tobacco Guangxi and its upstream and downstream industry chain partners, thereby facilitating innovative development and digital transformation within the company.

5.2 Opening and sharing mechanisms of the innovation chain

For China Tobacco Guangxi, in the process of digital transformation, the interaction relationship among the four types of innovation entities will change dynamically, and the participation of a large number of heterogeneous virtual entities makes it difficult to coordinate the interaction of innovation entities. Regarding the acquisition and utilization of innovation resources, the focus should not be on ownership but rather on the principle of "not seeking ownership, but seeking utility", aiming to obtain innovation resources across organizations[38]. The opening and sharing mechanisms aim to promote open cooperation and resource sharing among various innovation entities within the innovation chain.

5.2.1 Multi-dimensional resource sharing mechanisms. Innovation subjects in the innovation chain engage in open collaboration, encompassing technology transfer, joint R&D, and collaborative projects with various organizations, enterprises, and individuals, which involve technology transfer, joint R&D, joint innovation projects, and other forms. Innovation entities can leverage data sharing and collaborative mechanisms to jointly analyze and discover new innovation opportunities and solutions. Knowledge-sharing platforms facilitate the exchange of research findings, technical know-how, and innovative experiences. Resource-sharing platforms enable mutual assistance and complementary resource utilization, thereby enhancing efficiency and capabilities within the innovation chain. Innovation entities within the innovation chain establish data sharing and collaboration mechanisms to improve data availability and quality, providing a solid foundation for data-driven innovation.

Based on the lean concept, China Tobacco Guangxi has built an Internet innovation workshop to promote scientific management of the innovation process. By enriching innovation resources from the three dimensions of personnel, technology, and data, and through efficient collaborative toolkits, the communication efficiency of innovation teams is enhanced, response capabilities and agility are improved, and a one-stop solution for innovation activities is provided to enterprises. To foster an ecological innovation community promoting industry-commerce collaboration, a problem-oriented approach is adopted, encompassing micro-improvements, lean projects, QC group activities, and lean initiatives, forming repositories of achievements, case studies, methodologies, and expert resources.

5.2.2 Digital open innovation platform. Based on lean principles, China Tobacco Guangxi has created a digital innovation platform (Figure 3) to promote the scientific management of the innovation process.The platform aims to explore consumer databases, integrate data, form a closed business loop, and enhance synergies across these sectors, forms innovation synergy with synergistic empowerment, improves innovation efficiency with online tools, promotes innovation communication with interactive methods, accelerates innovation dissemination with community management, and standardizes innovation activities with visualization process, to create "six types of innovation", form "six libraries of innovation", and establish an "innovation pipeline". The platform facilitates the collective resolution of innovation challenges and amplifies the impact and influence of the innovation chain.

The opening and sharing mechanisms of the innovation chain can promote open cooperation, resource sharing, and collaborative innovation among innovation subjects, helping to break through information and resource barriers, providing an open innovation environment, and stimulating innovation vitality within the innovation chain.

5.3 Integration mechanisms of the industrial chain and innovation chain

Deploying the innovation chain around the industrial chain and highlighting industrial development is the main purpose of science and technology innovation, which can significantly enhance the overall effectiveness of China Tobacco Guangxi's digital innovation ecosystem.

5.3.1 Independent innovation strengthens the support system. As of December 2023, China Tobacco Guangxi has obtained a total of 1,030 valid patents, including 224 invention patents, 743 utility model patents, and 63 design patents. Additionally, the company has acquired 415 computer software copyright registrations and 482 domestic trademark registration certificates.China Tobacco Guangxi takes measures to strengthen the company's software development capabilities and ensure that the initiative and control of the company's core technologies and big data applications are firmly in its own hands. Relying on the company's big data and artificial intelligence laboratory platform, it has established the industry's independent software engineering R&D team within the system, which specializes in software front-end program development, back-end program development, database program development,

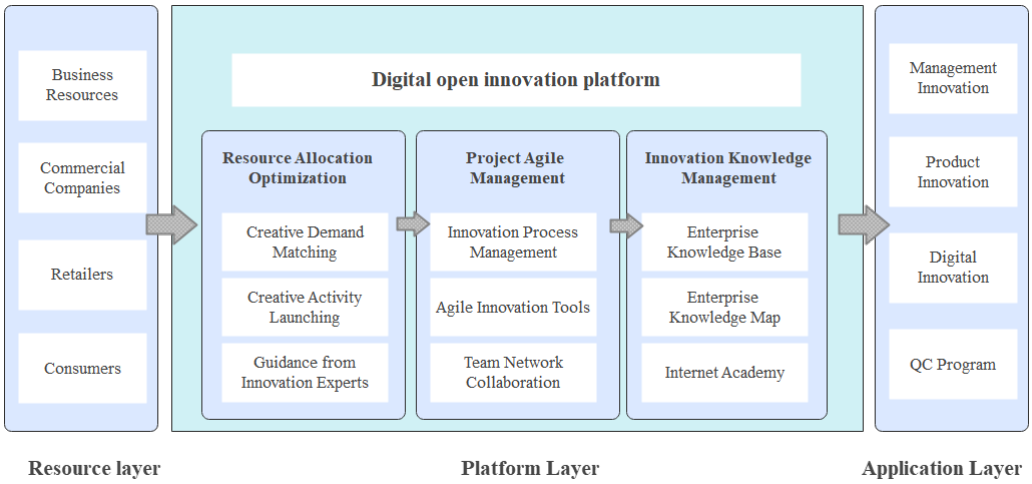

Figure 3: Digital open innovation platform of China Tobacco Guangxi.

big data algorithm design and implementation, technical research, etc. Through the establishment of daily morning classes, weekly work plans, weekly work reports, monthly performance appraisals, monthly technical exchange meetings, and other work systems, software engineers are assisted in improving self-management, stimulating intrinsic motivation among employees, and strengthening engineers' professional and technical capabilities. Through the preparation of the company's software self-development management standards, clear software development process management, team role division of labor and responsibilities, the establishment of software product requirements committee and technical committee respectively on the development of self-developed software products and technical feasibility of the development of the demand to ensure that the software development project is completed with high quality.

5.3.2 Digital innovation to create a dynamic system. China Tobacco Guangxi actively utilizes constantly updated data and drives the application of innovation in company development. Firstly, it leverages big data analysis to understand consumer patterns and R&D needs, integrating internal data with external urban, economic, demographic, and consumption metrics, to realize multi-source big data and fusion for the brand consumption market. Secondly, it carries out intelligent analysis research and application research of product placement based on cigarette consumption big data. Focusing on the core business needs of marketing, brand cultivation, and product placement at the tobacco industry end and cigarette ordering, brand cultivation, and product placement at the commercial end, combining the company's existing market big data and external big data resources, and utilizing technologies such as digital tagging, data mining, deep learning, and knowledge mapping, it will gain deep insights into the behavioral characteristics of cigarette consumption in different consumer groups and different consumption places, to build a big data tagging system for analysis of the correlation between cigarette products and consumption places. Third, it investigates multi-dimensional intelligent identification technologies for accessing cigarette product quality, encompassing image recognition, spectroscopy, and mass spectrometry, develop portable detection devices and intelligent algorithms, and compare genuine databases to achieve product authenticity evaluation. Fourth, promote the construction of a digital R&D platform for cigarette products. Utilizing various mathematical models and data, it can construct a digital design and maintenance platform specifically for its cigarette products, in order to realize the integration of cigarette product design and maintenance, quality monitoring, data analysis, and management.

5.3.3 Organizational innovation and upgrading capability system. Robust organizational change is a crucial guarantee to promote digital integration and innovation, China Tobacco Guangxi previously adopted a flexible and efficient organizational structure, including organizational backstage, middle stage, and front stage. Among them, the backstage is a strong digital foundation guarantee, including a digital innovation culture system, management system, thinking system, ecological system, methodology group, etc. The frontstage is a digital innovation project, connecting the market and the majority of users. The middle stage is the intermediate part connecting the backstage and the frontstage, based on the fertile soil of the backstage, forming a digital innovation capability system, continuously outputting innovative projects to the frontstage. The digital organization platform encompasses six major capabilities: research capability, R&D capability, operational capability, data capability, training capability, and management capability. These six capabilities consolidate common tools, data, and technologies in business operations, building a robust supporting platform that provides strong support for digital transformation.

6 Conclusion and Implications

Through a case study on the relationship mechanisms of the digital innovation ecosystem of China Tobacco Guangxi Company, this paper explores how an enterprise can integrate innovation management using digital technologies. It studies the collaborative innovation mechanisms, opening and sharing mechanisms, and integration mechanisms. It is found that effective relationship

mechanisms are the crucial element in constructing an enterprise innovation ecosystem, promoting continuous innovation and sustainable development within the enterprise.

The digital innovation ecosystem of China Tobacco Guangxi revolves around the mechanisms of collaborative innovation along the industrial chain. Through various approaches such as "Internet+" marketing collaborative innovation, collaborative production and manufacturing data, data-driven technological collaborative innovation, and collaborative construction of an innovative ecosystem within the industry, close cooperation and resource sharing with partners have been realized, enhancing the company's innovation capability and market competitiveness. In addition, the opening and sharing mechanisms of innovation chain provide an effective platform for the flow and exchange of internal and external innovation resources of China Tobacco Guangxi and enhance the innovation vitality of the enterprise through the data, knowledge, and resource-sharing mechanisms and the construction of digital open innovation platform. Finally, the integration mechanisms organically combine the innovation chain with the industrial chain, strengthen the support system through independent innovation, create a dynamic system through digital innovation, and upgrade the construction of the capability system through organizational innovation, forming a virtuous cycle that continues to promote the digital transformation and innovative development of China Tobacco Guangxi.

The study provides valuable insights and references for other enterprises seeking to establish a digital innovation ecosystem. First, enterprises should actively engage in collaborative innovation with partners and establish solid cooperative relationships to promote their innovative development. Second, the opening and sharing mechanisms is an important support for the digital innovation ecosystem. Enterprises should actively build an opening and sharing platform to promote the flow and exchange of internal and external innovation resources, thus enhancing innovation efficiency. Lastly, by focusing on the strategic placement of the innovation chain around the industrial chain and establishing integration mechanisms to promote the organic combination, enterprises should strengthen their capabilities for independent innovation, establish digital innovation platforms, enhance organizational capabilities for innovation upgrades, and promote a virtuous cycle of innovation and development.

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References

[1] Wei Jiang, Liu Jialing, and Liu Yang. New Trends and Problems of Innovation Strategy Theory in the Context of New Organization[J]. Journal of Management World, 2021, 37(7): 182-197,13.
[2] Rong Ke, Liu Xiein, Wei Jiang, and Hao Fei. Research on innovation ecosystems in the digital era[J]. Journal of Industrial Engineering and Engineering Management, 2023, (6):1-7.
[3] Hou Erxiu, Xu Rongqi, Yin Ximing, and Hou Wenli. Review of Research on Corporate Innovation Ecosystem Governance in the Digital Age[J]. Journal of Technology Economics, 2022, 41(11):78-93.
[4] Chen Kaihua. Accelerate the promotion of innovative development and digital transformation[J]. Outlook, 2020, (52): 24 -26.
[5] Adner R. Match Your Innovation Strategy to Your Innovation Ecosystem[J]. Harvard Business Review, 2006, 84(4):98-107.
[6] Wang Mengyu, Gu Fangjie. A study of the impact of corporate innovation performance based on the innovation ecosystem perspective[J]. Investment and Cooperation, 2024, (04): 155-157.
[7] Adner R, Kapoor R. Value creation in innovation ecosystems: how the structure of technological interdependence affects firm performance in new technology generations[J]. Strategic Management Journal, 2010, 31(3):306-333.
[8] Wu Jinxi. The connotation, characteristics and implications of innovation ecosystem[J]. Studies in Science of Science, 2014, 32(1):44-51, 91.
[9] Iansiti M, Levien R. Strategy as ecology[J]. Harvard Business Review, 2004, 82(3): 68-78.
[10] Zeng Guoping, Gou Youzhao, and Liu Lei. From innovation system to innovation ecosystem[J]. Studies in Science of Science, 2013, 31(1):4-12.
[11] Zhao Fang, Zeng Guoping. The Linkage Effects in Industrial Upgrading under Parallel Conditions of GVC and NVC-A Case Study on Shenzhen's Industrial Upgrading[J]. China Soft Science, 2014, (11):50-58.
[12] Liu Xielin, Sun Haiying, and Ma Xuemei. Management of Science and Technology from Innovation-Ecosystem Approach[J]. Science of Science and Management of S.& T., 2015, 36(1):18-27.
[13] Han Shaojie, Su Jingqin. Research on the construction of open innovation ecosystems of digital transformation enterprises-Theoretical foundation and future prospects[J]. Studies in Science of Science, 2023, 41(2):335-347.
[14] Llvonen L , Thalmann S , Manhart M ,et al. Reconciling digital transformation and knowledge protection: a research agenda[J]. Knowledge Management Research & Practice, 2018, 16(2):235-244.
[15] Zhang Chao, Chen Kaihua, and Mu Rongping. The digital innovation ecosystems: Theory building and a research agenda[J]. Science Research Management, 2021, 42(3):1-11.
[16] Sun Xinbo, Zhang Yuan, Wang Yongxia, and Sun Haobo. Digital Value Creation:Research Framework and Prospects[J]. Foreign Economics & Management, 2021, 43(10): 35-49.
[17] Pidun U, Reeves M, Wesselink E.How healthy is your business ecosystem?[J]. MIT Sloan Management Review, 2021, 62(3): 31-38.
[18] Wang P. Connecting the parts with the whole: Toward an information ecology theory of digital innovation ecosystems[J]. MIS Quarterly, 2021, 45(1B): 397-422.
[19] Suseno Y, Laurell C, Sick N. Assessing value creation in digital innovation ecosystems: A Social Media Analytics approach[J]. The Journal of Strategic Information Systems, 2018, 27(4):335 -349.
[20] Shan Zidan, Zeng Yanhong, Li Huimin, and Wang Yaqi. How Data Resources Reshape Multi-agent Co-opetition Relationships in Digital Innovation Ecosystem?-Deconstruction and Reorganization of Intelligent Driving Digital Innovation Ecosystem[J]. R&D Management, 2022, 34(6): 79-91.
[21] Adner R. Ecosystem as structure: An actionable construct for strategy[J]. Journal of Management,2017,43(1):39 -58.
[22] Buhe Chulu; Chen Ling. Digital Innovation Ecosystem:Concept,Structure and Operating Mechanism[J]. Forum on Science and Technology in China, 2022, (9): 54-62.
[23] Chae B K. A General Framework for Studying the Evolution of the Digital Innovation Ecosystem: the Case of Big Data[J]. International Journal of Information Management, 2019, 45: 83-94.
[24] Liu Yang, Dong Jiuyu, and Wei Jiang. The digital innovation management: Theoretical framework and future research[J]. Journal of Management World, 2020, 36(7): 198- 217.
[25] Yang Wei, Liu Jian, and Wu Jian. The impact of "population-flow" configuration on focal firm's performance-Empirical research on digital innovation ecosystems inartificial intelligence industry[J]. Studies in Science of Science, 2020, 38(11):2077-2086.
[26] Sawhney M, Verona G, Prandelli E. Collaborating to create: The Internet as a platform for customer engagement in product innovation[J]. Journal of Interactive Marketing, 2005, 19(4):4–17.
[27] Elia G, Margherita A, Passiante G. Digital entrepreneurship ecosystem: How digital technologies and collective intelligence are reshaping the entrepreneurial process[J]. Technological Forecasting and Social Change, 2020, 150: 119791.
[28] Hanelt A, Bohnsack R, Marz D, *et al.* A systematic review of the literature on digital transformation: Insights and implications for strategy and organizational change[J]. Journal of Management Studies, 2021, 58(5): 1159-1197.
[29] Chen Kaihua, Feng Ze, and Sun Qian. Big Data of Innovation, Innovation Governance Efficacy and Digital Transformation [J]. R&D Management, 2020, 32(6):1-12.
[30] Liang Yin, Zhao Yubo, and Liu Bing. Research on Monitoring And Early Warning of the Resilience of the Innovation Ecosystem in the China's National New Districts[J]. China Soft Science, 2020, (7): 92-111.
[31] Yang Wei, Lao Xiaoyun, Zhou Qing, and Zhang Luo. The governance niche configurations for the resilience of regional digital innovation ecosystem[J]. Studies in Science of Science, 2022, 40(3): 534-544.
[32] Yang Lihua, and Li Zhigang. A Concise Roadmap of Case Study Methods in Public Administration[J]. Journal of Zhengzhou University: Philosophy and Social Sciences Edition, 2022, 55 (04): 12-18+127..
[33] Yin R K. Case study research: Design and methods[M]. sage, 2009.
[34] Eisenhardt KM, Graebner M E. Theory building from cases: Opportunities and challenges[J]. Academy of Management Journal, 2007, 50(1): 25-32.

[35] Wang Hong, Zhang Zhongyi. Rebuilding of the value chain in tobacco industry from the perspective of "Supply by Order"[J]. Acta Tabacaria Sinica, 2006, 12(3): 11-16.

[36] Johnsen T, Ford D. Managing collaborative innovation in complex networks: Findings from exploratory interviews[C]. 16th Annual IMP Conference. Citeseer, 2000, 613-629.

[37] Tu Zhenzhou, Gu Xin. Study on the Synergetic Innovation Process of Industry-University-Research Institute Based on Knowledge Flow[J]. Studies in Science of Science, 2013, 31(9): 1381-1390.

[38] Zhang Haitao, Liu Qiyan. Research on the Ecological Architecture of Open Innovation Intellectual Resource: Taking the Technical Center of China Tobacco Yunnan industrial Company as an Example[J]. Knowledge Management Forum, 2017, (2): 167-174.